

Q2. (Marks: [8+1+1=10])

- (i) Derive the expression for energy of electron revolving around the nucleus using Bohr's model.
- (ii) Express it in terms of atomic units of energy, hartree, E_h .
- (iii) What is the energy (in E_h) of the electron in hydrogen atom according to Bohr's model?

Q3. (Marks: [3+2+5=10])

- (i) Write down the expression for time-independent Schrödinger equation by expanding the Hamiltonian operator, $\hat{H}$.
- (ii) State various conditions for well-behaved/acceptable wavefunctions.
- (iii) Show that $\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$ is an eigenfunction of the Hamiltonian operator $\hat{H}$ for a free-particle in x-dimension. What eigenvalue did you obtain?